

CO5 ASSESSMENT TOOL1

COURSE NAME: Database Management Systems

COURSE CODE: CSA0506

NAME: AAMEENA I

REG NO: 192511284

COURSE OUTCOME COVERED

CO5: Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability. (BL3, BL4)

Activity Tool : Mock Interview (Low)

Assessment Tool Weightage: 30%

What is the deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies

1. Interviewer: What is DBMS?

Candidate: DBMS stands for Database Management System. It is software used to store, manage, and retrieve data from a database.

2. Interviewer: What is deadlock in DBMS?

Candidate: Deadlock is a situation where two or more transactions wait for each other and cannot continue.

3. Interviewer: Why does deadlock occur?

Candidate: Deadlock occurs when transactions hold some resources and wait for resources held by other transactions.

4. Interviewer: What are the four conditions for deadlock?

Candidate: The four conditions are Mutual Exclusion, Hold and Wait, No Preemption, and Circular Wait.

5. Interviewer: What is Mutual Exclusion?

Candidate: It means a resource can be used by only one transaction at a time.

6. Interviewer: What is Hold and Wait?

Candidate: It means a transaction holds one resource while waiting for another resource.

7. Interviewer: What is No Preemption?

Candidate: It means a resource cannot be forcibly taken from a transaction. The transaction must release it.

8. Interviewer: What is Circular Wait?

Candidate: It means transactions form a cycle where each transaction waits for a resource held by another transaction.

9. Interviewer: How can we detect a deadlock?

Candidate: We can detect a deadlock using a wait-for graph. If the graph contains a cycle, a deadlock exists.

10 Interviewer: How can we prevent deadlock?

Candidate: Deadlock can be prevented by breaking any one of the four conditions, such as using resource ordering to prevent circular wait.



Code of Conduct:

I **AAMEENA I(192511284)**, certify that this submission and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Aameena I

